

HTML Lists, Tables, Forms & Semantics

Comprehensive Classroom Technical Handbook & Reference Guide

Module: HTML5 Core Prerequisites: Day 1-2 Fundamentals Target: Production Standards

Day 3 Learning Architecture

This session focuses on structural data presentation and user interaction pipelines in modern web applications.

01 Lists Structure	02 Data Tables	03 Forms & Inputs	04 Semantic Tags
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1. HTML Lists Deep Dive

 <dl>

HTML provides three distinct list elements tailored for specific content relationships: unordered groups, sequential steps, and key-value term definitions.

Unordered () Bulleted Used when item order is non-sequential. - HTML Syntax - CSS Styling - JavaScript Logic	Ordered () Indexed Used for sequential or ranked items. - HTML Structure - CSS Presentation - JS Behavior	Description (<dl>) Key-Value Pairs terms (<dt>) with definitions (<dd>). HTML Markup Language CSS Style Sheets
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List Syntax & Attributes Reference

```
<!-- Ordered List with Custom Start & Roman Type -->
<ol type="i" start="1">
  <li>HTML5 Markup</li>
  <li>CSS3 Styling</li>
</ol>

<!-- Description List for Metadata or Dictionaries -->
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language standard for web pages.</dd>
</dl>
```

2. HTML Data Tables & Grid Matrix

```
<table> <tr> <th> <td>
```

Tables structure complex multidimensional data. Proper table construction requires structural wrappers (`<thead>` , `<tbody>`) and cell spanning attributes (`colspan` , `rowspan`).

CLASS TIMETABLE MATRIX (DAY 3 REFACTORED IMPLEMENTATION)

Hours	Mon	Tue	Wed	Thu	Fri
Science	Maths	Art	English	Sport	
Science	Maths	Art	English	Sport	
LUNCH BREAK					
Science	Maths	Art	English	Sport	
Science	Maths	Art	English	Sport	
Science	Maths	Art	English (Extended Block)		

Cell Spanning Mechanics

ATTRIBUTE	FUNCTION
<code>rowspan="N"</code>	Merges a cell vertically across <i>N</i> rows down.
<code>colspan="N"</code>	Merges a cell horizontally across <i>N</i> columns right.

Table Accessibility Best Practices

- Always define a `<caption>` for context.
- Use `scope="col|row"` on `<th>` elements.
- Avoid using tables for page layout structure.

Timetable Code Snippet Structure

```
<table border="1">
  <caption>Weekly Timetable</caption>
  <tr>
    <th rowspan="7">Hours</th>
    <th>Mon</th><th>Tue</th><th>Wed</th><th>Thu</th><th>Fri</th>
  </tr>
  <tr><td colspan="5">Lunch</td></tr>
</table>
```

3. Interactive HTML Forms & Control Elements

`<form>` `<input>` `<button>`

Forms enable bidirectional interaction between users and web applications. Explicit label binding via matching `for` and `id` attributes ensures UI accessibility.

TAG / TYPE	CONTROL TYPE	KEY ATTRIBUTES & BEHAVIOR
<code>type="text"</code>	Single-line Text	<code>placeholder</code> , <code>required</code> , <code>maxlength</code>
<code>type="password"</code>	Masked Input	Obscures keystrokes for secure values.
<code>type="radio"</code>	Exclusive Choice	Grouped using identical <code>name</code> attributes.
<code>type="checkbox"</code>	Multi-select Toggle	Allows independent boolean selection.
<code><select></code>	Dropdown Menu	Contains nested <code><option></code> elements.
<code><textarea></code>	Multi-line Textbox	Defined by <code>rows</code> and <code>cols</code> properties.

Rendered Form Preview (Day 3 Exercise Clean UI Layout)

User Name

Password

Email Address

Select Gender

☒ Male ☐ Female

Select Skills

☒ HTML ☒ CSS ☐ JS

Select Course

HTML5 & Web Design

▼

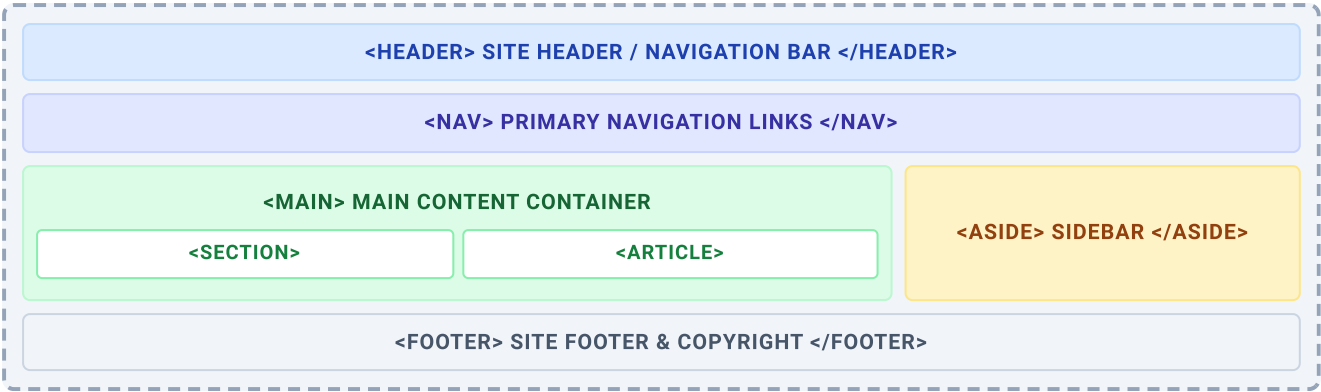
Note: Clicking radio/checkbox options functions directly in preview.

Submit Form

4. HTML5 Semantic Layout Architecture

<main> <header> <article>

Semantic tags replace generic `<div>` blocks to convey meaningful page structure to search engine crawlers, browsers, and screen readers.



SEMANTIC TAG	STRUCTURAL PURPOSE	BEST PRACTICE USAGE
<header>	Introductory content banner	Contains headings, logo, or navigation.
<nav>	Navigation menu block	Encapsulates primary site navigation links.
<main>	Dominant unique document body	Only one <main> allowed per page.
<article>	Self-contained composition	Re-usable content (blog post, news item).
<aside>	Tangentially related content	Sidebars, callout panels, related links.
<footer>	Document/section footer	Contains author info, legal terms, copyright.

Day 3 Code Cleanup & Syntax Rules Checklist

- Always match `<label for="id">` with `<input id="id">`.
- Group radio buttons using identical `name="..."` attributes.
- Ensure cell span counts in tables match total column matrix counts.
- Use lowercase element tags and quote all attribute values.